

SIMULTANEOUS VISIBILITY IN CUT-AND-PROJECT SETS

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ABSTRACT. We study simultaneous visibility in algebraic cut-and-project sets associated with real quadratic fields satisfying the Hammarhjelm condition. Given a finite subset S , we investigate the points that are visible from every point of S . Under a natural geometric condition on the window, we prove that the density of the simultaneous visibility set exists along every natural averaging sequence and derive an explicit formula for it. This extends previous results of Hammarhjelm on visible points from the origin in algebraic cut-and-project sets.

1. INTRODUCTION

Let $\Lambda \subset \mathbb{R}^d$ be a locally finite point set, and let $S \subset \Lambda$ be finite. We denote by $\Lambda_{\text{vis},S}$ the set of points in $\Lambda \setminus S$ that are visible from every point $\mathbf{s} \in S$. Geometrically, these are the points that can be reached simultaneously from all points of S by straight line segments whose interiors do not pass through any other point of Λ :

$$\Lambda_{\text{vis},S} = \{\mathbf{y} \in \Lambda \setminus S : \forall \mathbf{s} \in S, \forall t \in (0, 1), t\mathbf{s} + (1-t)\mathbf{y} \notin \Lambda\}.$$

Let $\mathcal{A} = \{A_T\}_{T \in \mathcal{I}}$ be an increasing sequence of sets whose union is $\mathbb{R}^d$, with index set $\mathcal{I} = \mathbb{N}$ or $\mathbb{R}$. The asymptotic density of Λ along $\mathcal{A}$ is defined by

$$\theta(\Lambda) \stackrel{\text{def}}{=} \lim_{T \rightarrow \infty} \frac{\#(\Lambda \cap A_T)}{\text{vol}(A_T)},$$

provided the limit exists.

The integer lattice $\Lambda = \mathbb{Z}^k$, with $k \geq 2$, is the classical example. The set $\mathbb{Z}_{\text{vis},\{\mathbf{0}\}}^k$ of lattice points visible from the origin is usually called the set of primitive points. It is well known that its density is

$$\theta(\mathbb{Z}_{\text{vis},\{\mathbf{0}\}}^k) = \frac{1}{\zeta(k)}$$

for standard averaging sequences, such as cubes and balls. See for example [BMP00], and see the appendix of [BMP00] for a discussion of more general averaging sets. It follows easily from the Chinese Remainder Theorem that the set $\mathbb{Z}^k$ contains arbitrarily large cubes containing no visible points from the origin (cf. [Apo76, Theorem 5.29] for the integer lattice, and [BMP00, Proposition 4] for arbitrary full-rank lattices). Thus, despite being uniformly discrete, $\mathbb{Z}_{\text{vis},\{\mathbf{0}\}}^k$ fails to be relatively dense and hence is not a Delone set (see Section 2.2). In fact, since these holes repeat periodically throughout the lattice, adjoining any set of zero density to $\mathbb{Z}_{\text{vis},\{\mathbf{0}\}}^k$ still does not produce a Delone set.

Denote by $\mathcal{P}$ the set of prime numbers. For each $p \in \mathcal{P}$, let

$$\pi_p : \mathbb{Z}^k \longrightarrow (\mathbb{Z}/p\mathbb{Z})^k, \quad (x_1, \dots, x_k) \longmapsto (x_1 \bmod p, \dots, x_k \bmod p)$$

be the coordinatewise projection. We write

$$s(p) \stackrel{\text{def}}{=} \#(\pi_p(S))$$

for the number of distinct residue classes modulo p represented by the elements of S .

Rumsey [Rum66, Theorem 7] proved that, with respect to the cubic averaging sequence $[-n, n]^k$, the density of $\mathbb{Z}_{\text{vis}, S}^k$ exists and is given by

$$\theta(\mathbb{Z}_{\text{vis}, S}^k) = \prod_{p \in \mathcal{P}} \left(1 - \frac{s(p)}{p^k}\right).$$

In the special case where the points of S are pairwise visible, this formula was proved earlier by Rearick [Rea66], with an error term. Rumsey extended the formula to certain infinite subsets S satisfying appropriate hypotheses and observed that the same density formula holds when averaging over rectangular boxes

$$[-n_1, n_1] \times \cdots \times [-n_k, n_k],$$

as $\min\{n_1, \dots, n_k\} \rightarrow \infty$. The corresponding density result for averaging over Euclidean balls follows from [KT26, Theorem 1.3]. We refer the reader to [BKP23; ACZ23; PH13; BMP00; Ara26] and the references therein for further details on simultaneous visibility and related results.

1.1. Main results. Consider the decomposition

$$\mathbb{R}^n = \mathbb{R}^d \oplus \mathbb{R}^m$$

for positive integers n, d, m . Let $\mathcal{L}$ be a lattice in $\mathbb{R}^n$, and denote by π_{phys} and π_{int} the projections onto the first d coordinates and the last m coordinates, respectively. Let $\mathcal{W} \subset \mathbb{R}^m$ be a bounded measurable subset, called the *window*. The associated cut-and-project set is defined by

$$\Lambda(\mathcal{W}, \mathcal{L}) \stackrel{\text{def}}{=} \{\pi_{\text{phys}}(\mathbf{x}) : \mathbf{x} \in \mathcal{L}, \pi_{\text{int}}(\mathbf{x}) \in \mathcal{W}\}. \quad (1)$$

We say that $\Lambda = \Lambda(\mathcal{W}, \mathcal{L})$ is *irreducible* if $\pi_{\text{int}}(\mathcal{L})$ is dense in $\mathbb{R}^m$, $\pi_{\text{phys}}|_{\mathcal{L}}$ is injective, and the boundary of the window has Lebesgue measure zero. The density of Λ along the averaging sequence

$$\{TD \stackrel{\text{def}}{=} \{t\mathbf{x} : \mathbf{x} \in D, t \in [0, T]\}\}_{T>0}$$

exists and is given by

$$\theta(\Lambda(\mathcal{W}, \mathcal{L})) = \frac{\text{vol}(\mathcal{W})}{\text{covol}(\mathcal{L})},$$

where D is any Jordan measurable set with $\text{vol}(D) > 0$, see for example, [MS14, Proposition 3.2]. In Section 2.2 we define cut-and-project sets in greater generality.

Marklof and Strömbergsson [MS15, Theorem 1] proved that, for any irreducible cut-and-project set (1), the density of $\Lambda_{\text{vis}, \{\mathbf{0}\}}$ exists and satisfies

$$0 < \theta(\Lambda_{\text{vis}, \{\mathbf{0}\}}) \leq \theta(\Lambda). \quad (2)$$

Hammarhjelm [Ham22] gave examples of cut-and-project sets $\Lambda(\mathcal{W}, \mathcal{L}) \subset \mathbb{R}^2$ for which the inequality in (2) is strict and computed $\theta(\Lambda_{\text{vis}, \{\mathbf{0}\}})$ explicitly. His constructions include the Ammann–Beenker point set and the vertex set of the Penrose tiling. Some of these examples were also considered by Sing [Sin]. These constructions are based on algebraic lattices over real quadratic fields.

To state our results, we first recall the necessary notions from algebraic number theory. Let $K = \mathbb{Q}(\sqrt{\mathfrak{d}})$ be a real quadratic field, where $\mathfrak{d} \geq 2$ is square-free, and let $\mathcal{O}_K$ be its

ring of integers. We assume throughout that $\mathcal{O}_K$ is a principal ideal domain. The norm of an element $x \in \mathcal{O}_K$ is defined by

$$N(x) = x\sigma(x),$$

where σ is the non-trivial Galois automorphism of K . The *unit group* $\mathcal{O}_K^\times$ consists of the elements $x \in \mathcal{O}_K$ such that $N(x) = \pm 1$. The unique element $\lambda = \lambda_K \in \mathcal{O}_K^\times$ such that

$$\mathcal{O}_K^\times = \{\pm 1\} \times \{\lambda^i : i \in \mathbb{Z}\}, \quad \lambda > 1,$$

is called the *fundamental unit* of K . Let $I \subseteq \mathcal{O}_K$ be a non-trivial ideal. The *norm* of I is defined by

$$N(I) \stackrel{\text{def}}{=} \#(\mathcal{O}_K/I),$$

which is always finite. Moreover, $N(I) = |N(x)|$, where I is generated by x . The Dedekind zeta function of K is then defined by

$$\zeta_{\mathcal{O}_K}(s) \stackrel{\text{def}}{=} \sum_{\mathbf{0} \neq I \subseteq \mathcal{O}_K} \frac{1}{N(I)^s}, \quad \Re(s) > 1.$$

We refer the reader to [Mar18] for further details.

The Minkowski embedding

$$f : K \hookrightarrow \mathbb{R}^2, \quad \alpha \mapsto (\alpha, \sigma(\alpha)),$$

embeds $\mathcal{O}_K$ as a lattice $f(\mathcal{O}_K)$ in $\mathbb{R}^2$.

Let $\lambda > 1$ denote the fundamental unit of $\mathcal{O}_K$. We say that K satisfies the *Hammarhjelm condition* if

$$f(\mathcal{O}_K) \cap ((1, \lambda) \times [-1, 1]) = \emptyset.$$

For $2 \leq \mathfrak{d} \leq 100$, the Hammarhjelm condition is satisfied only for

$$\mathfrak{d} = 2, 5, 13, 29, 53;$$

see [GKW25; Ham22]. Recently, Rudnick [Rud26] completed the classification by proving that the real quadratic fields satisfying the Hammarhjelm condition are precisely those with

$$\mathfrak{d} = 2, 5, 13, 29, 53, 173, 293.$$

In Lemma 3.2, we strengthen the Hammarhjelm condition by proving that, for each of these fields,

$$f(\mathcal{O}_K) \cap \left((1, \lambda) \times \left[-\left(1 + \frac{1}{\lambda}\right), \left(1 + \frac{1}{\lambda}\right) \right] \right) = \emptyset,$$

which is crucial for the geometric condition on the window in Theorem 1.1.

By abuse of notation, for $\mathbf{x} = (x_1, \dots, x_d) \in K^d$ we write

$$\sigma(\mathbf{x}) \stackrel{\text{def}}{=} (\sigma(x_1), \dots, \sigma(x_d)).$$

For every $d \in \mathbb{N}$,

$$\mathcal{L} \stackrel{\text{def}}{=} \{(\mathbf{x}, \sigma(\mathbf{x})) : \mathbf{x} \in \mathcal{O}_K^d\} \tag{3}$$

is a lattice in $\mathbb{R}^{2d}$. Let $d \geq 2$, let $\mathcal{L}$ be the lattice in (3), and $\mathcal{W} \subset \mathbb{R}^d$ be a convex, centrally symmetric window. Suppose that the real quadratic field K satisfies the Hammarhjelm condition. For the associated cut-and-project set $\Lambda = \Lambda(\mathcal{W}, \mathcal{L})$, the density of the set of visible points from the origin, along the averaging sets TD , is given by

$$\theta(\Lambda_{\text{vis}, \{\mathbf{0}\}}) = \left(1 - \frac{1}{\lambda^d}\right) \frac{\theta(\Lambda)}{\zeta_{\mathcal{O}_K}(d)},$$

where D is any Jordan measurable set with $\text{vol}(D) > 0$. See [Ham22, Theorems 4.9, 4.12, 4.15] and [GKW25, Theorem 5.1] for an effective result. Moreover, [GKW25, Theorem 1.1] implies that $\Lambda_{\text{vis},\{0\}}$ contains arbitrarily large holes. In fact, it follows from the proof of [KO26, Theorem 1.3], together with [HuRi15, Proposition 2.12], that these holes repeat. Consequently, adjoining any set of zero density to $\Lambda_{\text{vis},\{0\}}$ still does not produce a Delone set.

Let $S \subset \Lambda$ be finite, and let $\mathcal{P}_{\mathcal{O}_K}$ be the set of nonzero prime ideals of $\mathcal{O}_K$. For each $\mathfrak{p} \in \mathcal{P}_{\mathcal{O}_K}$, let

$$\pi_{\mathfrak{p}} : \mathcal{O}_K^d \rightarrow (\mathcal{O}_K/\mathfrak{p})^d$$

denote the natural projection, and define

$$s(\mathfrak{p}) \stackrel{\text{def}}{=} \#(\pi_{\mathfrak{p}}(S)).$$

For $\mathbf{s} \in \Lambda$, define

$$\mathcal{W}_{\mathbf{s}} \stackrel{\text{def}}{=} \frac{1}{\lambda}(\mathcal{W} + (\lambda + 1)\sigma(\mathbf{s})). \quad (4)$$

If $I \subset \Lambda$ is nonempty and finite, define

$$\mathcal{W}_I \stackrel{\text{def}}{=} \bigcap_{\mathbf{s} \in I} \mathcal{W}_{\mathbf{s}}.$$

Suppose that $\mathcal{W}$ is convex, then each $\mathcal{W}_I$ is convex. Furthermore, if we assume that $\mathcal{W}$ is bounded and centrally symmetric, and that the points of the finite set S satisfy the geometric condition that, for every $\mathbf{s} \in S$,

$$-(\mathcal{W} - \sigma(\mathbf{s})) \subset \left(1 + \frac{1}{\lambda}\right)(\mathcal{W} - \sigma(\mathbf{s})), \quad (5)$$

then (see Lemma 4.1)

$$\mathcal{W}_{\mathbf{s}} \subset \mathcal{W}.$$

In the following theorem, we prove that the density of the set of points simultaneously visible from a finite set S along $\{TD\}_{T>0}$ exists and derive an explicit formula for it, under the assumption that each $\mathbf{s} \in S$ satisfies the geometric condition (5).

Theorem 1.1. *Let $K = \mathbb{Q}(\sqrt{d})$ be a real quadratic field with ring of integers $\mathcal{O}_K$ satisfying the Hammarhjelm condition. Let $d \geq 2$, let $\mathcal{L}$ be the lattice defined in (3), and let $\mathcal{W} \subset \mathbb{R}^d$ be a bounded, convex, centrally symmetric window. Suppose that $\Lambda = \Lambda(\mathcal{W}, \mathcal{L})$ is the associated cut-and-project set. Let $S \subset \Lambda$ be finite and assume that (5) holds for every $\mathbf{s} \in S$. Then the density of $\Lambda_{\text{vis},S}$ along $\{TD\}_{T>0}$ exists and is given by*

$$\theta(\Lambda_{\text{vis},S}) = \text{dens}(\mathcal{L}) \left(\text{vol}(\mathcal{W}) - \sum_{\emptyset \neq I \subseteq S} (-1)^{\#(I)+1} \text{vol}(\mathcal{W}_I) \right) \prod_{\mathfrak{p} \in \mathcal{P}_{\mathcal{O}_K}} \left(1 - \frac{s(\mathfrak{p})}{N(\mathfrak{p})^d} \right).$$

Here $D \subset \mathbb{R}^d$ is any bounded convex set with $\text{vol}(D) > 0$.

It follows from [KO26, Theorem 1.1] that if the sets $\mathbb{Z}_{\text{vis},S}^k$ have positive density along $[-L, L]^k$, then the set $\mathbb{Z}_{\text{vis},S}^k$ is a weak model set of maximal density (see Section 2.2 for the definition). Furthermore, it follows from the proof of [KO26, Theorem 1.3] that the set $\Lambda_{\text{vis},\{0\}}$ is also a weak model set of maximal density along the sequence $\{TD\}_{T>0}$. In the following corollary, we extended this result to the set $\Lambda_{\text{vis},S}$.

Corollary 1.2. *Under the hypotheses of Theorem 1.1, suppose that the density $\theta(\Lambda_{\text{vis},S})$ is positive along $\{TD\}_{T>0}$. Then the set $\Lambda_{\text{vis},S}$ is a weak model set of maximal density.*

Remark 1.3. By [BHS17, Theorem 7] and Corollary 1.2, the diffraction spectrum of $\Lambda_{\text{vis},S}$ is pure point and translation-bounded.

It follows from the proof of Corollary 1.2, together with [HuRi15, Proposition 2.12], that the set $\Lambda_{\text{vis},S}$ contains arbitrarily large holes, and these holes repeat. Consequently, adjoining any set of zero density to $\Lambda_{\text{vis},S}$ still does not produce a Delone set.

2. PRELIMINARIES

2.1. Averaging sequences. Let G be a σ -compact locally compact Abelian group equipped with Haar measure θ_G . A sequence $\{A_T\}_{T \in \mathcal{I}}$ of compact measurable subsets of G is an *averaging sequence* if

- (1) for $T < T'$, $A_T \subseteq A_{T'}$,
- (2) for every T , $\theta_G(A_T) < \infty$, and
- (3) $\bigcup_{T \in \mathcal{I}} A_T = G$.

An averaging sequence is a *van Hove sequence* if $\theta_G(A_T) > 0$ for all sufficiently large T , and for every compact set $K \subseteq G$, the K -boundary is asymptotically negligible, that is,

$$\lim_{T \rightarrow \infty} \frac{\theta_G(\partial^K A_T)}{\theta_G(A_T)} = 0,$$

where

$$\partial^K A_T \stackrel{\text{def}}{=} ((K + A_T) \setminus A_T^\circ) \cup ((-K + \overline{A_T}) \cap A_T)$$

is the K -boundary of A_T . By A_T^c , A_T° and $\overline{A_T}$ denote the complement, interior and closure of $A_T \subset G$, respectively, see [Moo02, p. 125] and [Sch98, p. 145]. The existence of van Hove sequences in σ -compact LCAGs is shown in [Sch98]. In the Euclidean setting, namely $G = \mathbb{R}^d$, standard examples of van Hove sequences include increasing sequences of Euclidean balls and cubes. More generally, dilates TA of a Jordan set $A \subset \mathbb{R}^d$ with nonempty interior:

Lemma 2.1. *Let $A \subset \mathbb{R}^d$ be bounded and measurable, with nonempty interior and*

$$\text{vol}(\partial A) = 0.$$

Then the family $\{TA\}_{T>0}$ forms a van Hove sequence.

Proof. We have $\bigcup_{T>0} TA = \mathbb{R}^d$, the family TA is clearly nested, and $0 < \text{vol}(A) < \infty$. Since A is bounded, ∂A is a compact set and the volume of its ε -neighborhoods tend to zero, that is

$$\text{vol}(\{\mathbf{w} \in \mathbb{R}^d : \exists \mathbf{z} \in \partial A, \mathbf{w} \in B_\varepsilon(\mathbf{z})\}) \xrightarrow{\varepsilon \rightarrow 0} \text{vol}(\partial A) = 0.$$

Given a compact set K , let $R > 0$ be the radius of an Euclidean ball with center at the origin that contains K . The K -boundary of the dilated set TA is contained in the R -neighborhood of the boundary of TA , namely

$$\partial^K(TA) \subset \{\mathbf{y} \in \mathbb{R}^d : \exists \mathbf{x} \in \partial(TA), \mathbf{y} \in B_R(\mathbf{x})\}.$$

Since $T(\partial A) = \partial(TA)$, we have

$$\begin{aligned} \frac{\text{vol}(\partial^K(TA))}{\text{vol}(TA)} &\leq \frac{\text{vol}(\{\mathbf{y} \in \mathbb{R}^d : \exists \mathbf{x} \in \partial(TA), \mathbf{y} \in B_R(\mathbf{x})\})}{T^d \text{vol}(A)} \\ &= \frac{T^d \text{vol}(\{\mathbf{w} \in \mathbb{R}^d : \exists \mathbf{z} \in \partial A, \mathbf{w} \in B_{R/T}(\mathbf{z})\})}{T^d \text{vol}(A)} \xrightarrow{T \rightarrow \infty} 0. \end{aligned}$$

□

We shall use the fact that every bounded convex subset of $\mathbb{R}^d$ has boundary of Lebesgue measure zero, thus is Jordan measurable [Sza97].

The upper and lower densities of $\Lambda \subset G$ with respect to an averaging sequence $\{A_T\}_{T \in \mathcal{I}}$ are defined by

$$\bar{\theta}(\Lambda) \stackrel{\text{def}}{=} \limsup_{T \rightarrow \infty} \frac{\#(\Lambda \cap A_T)}{\text{vol}(A_T)} \quad \text{and} \quad \underline{\theta}(\Lambda) \stackrel{\text{def}}{=} \liminf_{T \rightarrow \infty} \frac{\#(\Lambda \cap A_T)}{\text{vol}(A_T)},$$

respectively.

2.2. General cut-and-project sets. Let G and H be σ -compact, locally compact Abelian groups (LCAGs) equipped with Haar measures θ_G and θ_H , respectively. Throughout, we assume that H is compactly generated. By the Structure Theorem for compactly generated locally compact Abelian groups, H is topologically isomorphic to $\mathbb{R}^d \times \mathbb{Z}^n \times \mathbb{K}$, for some integers $d, n \geq 0$ and some compact Abelian group $\mathbb{K}$, see [HeRo79, Theorem 9.8].

We refer to G as the *physical space* and H as the *internal space*. Let $\mathcal{L} \subset G \times H$ be a lattice, and let $\mathcal{W} \subset H$ be a measurable window. Let

$$\pi_{\text{phys}} : G \times H \rightarrow G, \quad \pi_{\text{int}} : G \times H \rightarrow H$$

denote the natural projections. We assume that the restriction $\pi_{\text{phys}}|_{\mathcal{L}} : \mathcal{L} \rightarrow G$ is injective and that the image $\pi_{\text{int}}(\mathcal{L})$ is dense in H . The associated cut-and-project set is

$$\Lambda(\mathcal{W}, \mathcal{L}) \stackrel{\text{def}}{=} \{\pi_{\text{phys}}(z) \in G : z \in \mathcal{L}, \pi_{\text{int}}(z) \in \mathcal{W}\}.$$

The cut-and-project set $\Lambda(\mathcal{W}, \mathcal{L})$ is called a *model set* if the window $\mathcal{W}$ is relatively compact and has non-empty interior, that is,

$$\mathcal{W}^\circ \neq \emptyset,$$

see [BHS17] and the references therein. If, in addition, $\mathcal{W} = \overline{\mathcal{W}^\circ}$ and $\mathcal{W}$ is compact, then the window is called *proper*. If $\mathcal{W}$ is a proper window, has non-empty interior, and satisfies

$$\theta_H(\partial \mathcal{W}) = 0,$$

then the cut-and-project set $\Lambda(\mathcal{W}, \mathcal{L})$ is called *irreducible*.

The density of an irreducible cut-and-project set $\Lambda(\mathcal{W}, \mathcal{L})$ along a van Hove sequence $\mathcal{A} = (A_n)_{n \in \mathbb{N}}$ exists and is given by

$$\theta(\Lambda(\mathcal{W}, \mathcal{L})) = \lim_{n \rightarrow \infty} \frac{\#(\Lambda(\mathcal{W}, \mathcal{L}) \cap A_n)}{\theta_G(A_n)} = \theta(\mathcal{L}) \cdot \theta_H(\mathcal{W}), \quad (6)$$

see [Moo02, Theorem 1], [Sch98], [HuRi15, Lemma 3.2, Remark 3.3], [MS14, Proposition 3.2], and the references therein. Moreover, the density of irreducible cut-and-project sets is independent of the van Hove sequence, see [HuRi15, Remark 3.3(i)]. Irreducible cut-and-project sets are Delone sets, see [HuRi15, p. 744].

Recall that a set $\mathcal{D} \subset G$ is *uniformly discrete* if there exists an open neighborhood U of the identity in G such that any of its translates xU , with $x \in G$, contains at most one point of $\mathcal{D}$. The set $\mathcal{D} \subset G$ is *relatively dense* if there exists a compact set $K \subset G$ such that $\mathcal{D}K = G$. A subset $\mathcal{D} \subset G$ is called a *Delone set* if it is both uniformly discrete and relatively dense.

In the main results of this paper, the windows $\mathcal{W} \subset \mathbb{R}^m$ are assumed to be bounded, convex, and centrally symmetric. Here centrally symmetric means that

$$\mathcal{W} = -\mathcal{W}.$$

We say that $\Lambda(\mathcal{W}, \mathcal{L})$ is a *weak model set of maximal density* relative to a given averaging sequence $\mathcal{A}$ if $\Lambda(\mathcal{W}, \mathcal{L})$ is a weak model set, and if the density of $\Lambda(\mathcal{W}, \mathcal{L})$ relative to $\mathcal{A}$ exists and is given by

$$\theta(\Lambda(\mathcal{W}, \mathcal{L})) = \theta(\mathcal{L}) \cdot \theta_H(\overline{\mathcal{W}}).$$

2.3. Visibility from a point in a cut-and-project set. Let $K = \mathbb{Q}(\sqrt{\mathfrak{d}})$ be a real quadratic field with ring of integers $\mathcal{O}_K$ satisfying the Hammarhjelm condition. For two distinct points $\mathbf{x} = (x_1, \dots, x_k)$ and $\mathbf{y} = (y_1, \dots, y_k)$ in $\mathcal{O}_K^k$, we define

$$\gcd(\mathbf{x} - \mathbf{y}) = \gcd((x_1 - y_1), \dots, (x_k - y_k)),$$

where $(x_i - y_i)$ denotes the ideal generated by $x_i - y_i$. Since $\mathcal{O}_K$ is a principal ideal domain, every such ideal is principal. By abuse of notation, we write $\gcd(\mathbf{x} - \mathbf{y})$ for a generator of this ideal. In particular, if

$$\gcd(\mathbf{x} - \mathbf{y}) = \mathcal{O}_K,$$

we simply write

$$\gcd(\mathbf{x} - \mathbf{y}) = 1.$$

We also define

$$\gcd(\mathbf{0}) \stackrel{\text{def}}{=} 0.$$

Let $\mathcal{W} \subset \mathbb{R}^d$ be a bounded, convex, centrally symmetric window, and let $\mathcal{L}$ be the lattice defined in (3). By [GKW25, Lemma 5.6],

$$\Lambda_{\text{vis}, \{\mathbf{0}\}}(\mathcal{W}, \mathcal{L}) = \left\{ \mathbf{x} \in \Lambda(\mathcal{W}, \mathcal{L}) \setminus \{\mathbf{0}\} : \gcd(\mathbf{x}) = 1, \sigma(\mathbf{x}) \notin \frac{1}{\lambda} \mathcal{W} \right\}.$$

The following lemma extends this characterization to visibility from an arbitrary point $\mathbf{s} \in \Lambda(\mathcal{W}, \mathcal{L})$ satisfying a suitable geometric condition.

Lemma 2.2. *Let $K = \mathbb{Q}(\sqrt{\mathfrak{d}})$ be a real quadratic field with ring of integers $\mathcal{O}_K$ satisfying the Hammarhjelm condition. Let $\mathcal{W} \subset \mathbb{R}^d$ be a bounded, convex, centrally symmetric window, let $\mathcal{L}$ be the lattice defined in (3), and let $\Lambda = \Lambda(\mathcal{W}, \mathcal{L})$ be the associated cut-and-project set. Let $\mathbf{s} \in \Lambda$ be such that*

$$-(\mathcal{W} - \sigma(\mathbf{s})) \subset \left(1 + \frac{1}{\lambda}\right) (\mathcal{W} - \sigma(\mathbf{s})).$$

Then the set of points visible from $\mathbf{s}$ is given by

$$\Lambda_{\text{vis}, \{\mathbf{s}\}} = \left\{ \mathbf{x} \in \Lambda : \gcd(\mathbf{x} - \mathbf{s}) = 1, \sigma\left(\frac{\mathbf{x} - \mathbf{s}}{\lambda}\right) \notin \mathcal{W} - \sigma(\mathbf{s}) \right\}. \quad (7)$$

3. PROOF OF LEMMA 2.2

Let $K = \mathbb{Q}(\sqrt{\mathfrak{d}})$ be a real quadratic field satisfying the Hammarhjelm condition, and let $\lambda > 1$ be its fundamental unit. Rudnick [Rud26] shows that

$$\mathfrak{d} \in \{2, 5, 13, 29, 53, 173, 293\}.$$

Observe that if

$$\mathfrak{d} \in \{13, 29, 53, 173, 293\},$$

then

$$\mathfrak{d} = x^2 + 4, \quad x \in \{3, 5, 7, 13, 17\}.$$

Writing $x = 2m + 1$, we have

$$\mathcal{O}_K = \mathbb{Z}[\omega], \quad \omega = \frac{1 + \sqrt{\mathfrak{d}}}{2},$$

and

$$\lambda = \frac{x + \sqrt{\mathfrak{d}}}{2} = m + \omega.$$

In particular,

$$N(\lambda) = -1, \quad \sigma(\lambda) = -\frac{1}{\lambda},$$

and

$$\frac{1}{\lambda} = \frac{\sqrt{\mathfrak{d}} - x}{2}, \quad 1 + \frac{1}{\lambda} = \frac{\sqrt{\mathfrak{d}} - x + 2}{2}.$$

Lemma 3.1. *Let $K = \mathbb{Q}(\sqrt{\mathfrak{d}})$ be a real quadratic field satisfying the Hammarhjelm condition, and let λ be its fundamental unit. Then*

$$1 + \frac{1}{\lambda} \leq \lambda.$$

Equality occurs only for $K = \mathbb{Q}(\sqrt{5})$.

Proof. By the discussion at the beginning of Section 3, it suffices to consider

$$\mathfrak{d} \in \{2, 5, 13, 29, 53, 173, 293\}.$$

For $\mathfrak{d} = 2$ one has $\lambda = 1 + \sqrt{2}$, and for $\mathfrak{d} = 5$ one has $\lambda = (1 + \sqrt{5})/2$. Thus the desired inequality is clear in these two cases, with equality only when $\mathfrak{d} = 5$.

For $\mathfrak{d} \in \{13, 29, 53, 173, 293\}$, we have

$$\lambda = \frac{x + \sqrt{\mathfrak{d}}}{2}, \quad \mathfrak{d} = x^2 + 4, \quad x \in \{3, 5, 7, 13, 17\}.$$

Since $\sqrt{\mathfrak{d}} > x$, we have $\lambda > x \geq 3$, and hence

$$1 + \frac{1}{\lambda} < 2 < \lambda.$$

□

Denote

$$\mathbb{P} \stackrel{\text{def}}{=} \{\pi \in \mathcal{O}_K : \pi \text{ is prime and } 1 < \pi < \lambda\}.$$

The set $\mathbb{P}$ plays the role of a fundamental domain for the action of the unit group on prime elements of $\mathcal{O}_K$: after multiplying by a unit, and possibly by -1 , every prime element has a representative in $\mathbb{P}$.

It follows from [Ham22, Lemma 4.7] that, if $\mathfrak{d} = 2$, then

$$|\sigma(\pi)| \geq 1 + \frac{1}{\lambda} \quad \text{for all } \pi \in \mathbb{P},$$

and if $\mathfrak{d} = 5$, then [Ham22, Lemma 4.13] gives the stronger bound

$$|\sigma(\pi)| \geq 2\lambda \quad \text{for all } \pi \in \mathbb{P}.$$

Lemma 3.2. *Let $K = \mathbb{Q}(\sqrt{\mathfrak{d}})$ be a real quadratic field satisfying the Hammarhjelm condition. Then, for every $\pi \in \mathbb{P}$,*

$$|\sigma(\pi)| \geq 1 + \frac{1}{\lambda}.$$

Moreover, for $\mathfrak{d} \in \{2, 13, 29, 53, 173, 293\}$, this bound is sharp.

Proof. By [Ham22, Lemmas 4.7 and 4.13], the desired lower bound holds for $\mathfrak{d} = 2, 5$. Moreover, for $\mathfrak{d} = 2$, the prime element $\pi = \sqrt{2}$ satisfies

$$1 < \sqrt{2} < \lambda, \quad |\sigma(\sqrt{2})| = \sqrt{2} = 1 + \frac{1}{\lambda},$$

so the bound is sharp in this case. It remains to treat

$$\mathfrak{d} \in \{13, 29, 53, 173, 293\}.$$

As discussed at the beginning of Section 3, there is $x \in \{3, 5, 7, 13, 17\}$ such that

$$\mathfrak{d} = x^2 + 4.$$

Writing $x = 2m + 1$, we have

$$\mathcal{O}_K = \mathbb{Z}[\omega], \quad \omega = \frac{1 + \sqrt{\mathfrak{d}}}{2}, \quad \lambda = m + \omega.$$

Also,

$$\sigma(\lambda) = -\frac{1}{\lambda}, \quad \sigma(\omega) = -m - \frac{1}{\lambda}.$$

Let $\pi \in \mathbb{P}$. There exist unique integers $a, b \in \mathbb{Z}$ such that

$$\pi = a + b\omega.$$

Since $\omega - \sigma(\omega) = \sqrt{\mathfrak{d}}$, we have

$$\sigma(\pi) = \pi - b\sqrt{\mathfrak{d}}. \tag{8}$$

We divide the proof into cases according to b .

- If $b \leq -1$, then by (8),

$$\sigma(\pi) = \pi + |b|\sqrt{\mathfrak{d}} > 1 + \sqrt{\mathfrak{d}}.$$

Since $1 + \lambda^{-1} < 2 < \sqrt{\mathfrak{d}}$, it follows that

$$|\sigma(\pi)| > 1 + \frac{1}{\lambda}.$$

- If $b = 0$, then $\pi = a \in \mathbb{Z}$. Since $\pi > 1$, we have $a \geq 2$, and hence

$$|\sigma(\pi)| = a \geq 2 > 1 + \frac{1}{\lambda}.$$

- If $b \geq 2$, then by (8),

$$\sigma(\pi) = \pi - b\sqrt{\mathfrak{d}} \leq \pi - 2\sqrt{\mathfrak{d}}.$$

Since $\mathfrak{d} = x^2 + 4$, we have $x < \sqrt{\mathfrak{d}}$, and therefore

$$\lambda = \frac{x + \sqrt{\mathfrak{d}}}{2} < \sqrt{\mathfrak{d}}.$$

As $\pi < \lambda$, we get

$$\sigma(\pi) < \sqrt{\mathfrak{d}} - 2\sqrt{\mathfrak{d}} = -\sqrt{\mathfrak{d}}.$$

Since $\sqrt{\mathfrak{d}} > 2 > 1 + \lambda^{-1}$, it follows that

$$|\sigma(\pi)| > 1 + \frac{1}{\lambda}.$$

- If $b = 1$, then $\pi = a + \omega$. The inequality $\pi < \lambda = m + \omega$ gives $a \leq m - 1$. Hence

$$\sigma(\pi) = a + \sigma(\omega) = a - m - \frac{1}{\lambda} \leq -1 - \frac{1}{\lambda}.$$

Thus

$$|\sigma(\pi)| \geq 1 + \frac{1}{\lambda}.$$

It remains to prove sharpness for

$$\mathfrak{d} \in \{13, 29, 53, 173, 293\}.$$

Set

$$\pi_0 = \lambda - 1.$$

Since $\lambda > x \geq 3$, we have

$$1 < \pi_0 < \lambda.$$

Moreover,

$$|\sigma(\pi_0)| = |\sigma(\lambda) - 1| = \left| -\frac{1}{\lambda} - 1 \right| = 1 + \frac{1}{\lambda}.$$

Finally,

$$N_{K/\mathbb{Q}}(\lambda - 1) = -x.$$

Since $x \in \{3, 5, 7, 13, 17\}$ is a rational prime, the ideal $(\lambda - 1)$ has prime norm. Hence $(\lambda - 1)$ is a prime ideal, so $\lambda - 1$ is a prime element. Therefore $\pi_0 \in \mathbb{P}$, and the bound is sharp. $\square$

Proof of Lemma 2.2. We first prove the inclusion $\subseteq$ in (7). Let $\mathbf{x} \in \Lambda_{\text{vis}, \{\mathbf{s}\}}$. We show that

$$\gcd(\mathbf{x} - \mathbf{s}) = 1 \quad \text{and} \quad \sigma\left(\frac{\mathbf{x} - \mathbf{s}}{\lambda}\right) \notin \mathcal{W} - \sigma(\mathbf{s}).$$

Suppose first that

$$\sigma\left(\frac{\mathbf{x} - \mathbf{s}}{\lambda}\right) \in \mathcal{W} - \sigma(\mathbf{s}).$$

Then

$$\mathbf{y}' = \mathbf{s} + \frac{\mathbf{x} - \mathbf{s}}{\lambda}$$

belongs to Λ , and since $\lambda > 1$, the point $\mathbf{y}'$ lies in the interior of the line segment joining $\mathbf{s}$ and $\mathbf{x}$. This contradicts the visibility of $\mathbf{x}$ from $\mathbf{s}$.

Next, suppose that $\gcd(\mathbf{x} - \mathbf{s}) \neq 1$. Then there exists a prime element $\pi \in \mathcal{O}_K$ dividing $\mathbf{x} - \mathbf{s}$. After multiplying π by a unit, and possibly by -1 , we may assume that $\pi \in \mathbb{P}$. Set

$$\mathbf{y} = \mathbf{s} + \frac{\mathbf{x} - \mathbf{s}}{\pi}.$$

Then $\mathbf{y} \in \mathcal{O}_K^d$, and since $1 < \pi < \lambda$, the point $\mathbf{y}$ lies in the interior of the line segment joining $\mathbf{s}$ and $\mathbf{x}$. It remains to show that $\mathbf{y} \in \Lambda(\mathcal{W}, \mathcal{L})$. We have

$$\sigma(\mathbf{y}) = \sigma(\mathbf{s}) + \sigma(\pi)^{-1} \sigma(\mathbf{x} - \mathbf{s}),$$

so it suffices to prove that $\sigma(\mathbf{y}) \in \mathcal{W}$.

Case 1. Assume $\sigma(\pi) > 0$.

By Lemma 3.2,

$$\sigma(\pi) \geq 1 + \frac{1}{\lambda},$$

and hence $0 < \sigma(\pi)^{-1} < 1$. Thus $\sigma(\mathbf{y})$ is a convex combination of $\sigma(\mathbf{s})$ and $\sigma(\mathbf{x})$. Since $\sigma(\mathbf{s}), \sigma(\mathbf{x}) \in \mathcal{W}$, the convexity of $\mathcal{W}$ implies that $\sigma(\mathbf{y}) \in \mathcal{W}$.

Case 2. Assume $\sigma(\pi) < 0$.

By Lemma 3.2, $\sigma(\pi) \leq -1 - \frac{1}{\lambda}$. Set

$$\mathbf{v} = \sigma(\mathbf{x}) - \sigma(\mathbf{s}), \quad c = \sigma(\pi)^{-1} \in \left[-\frac{\lambda}{1 + \lambda}, 0 \right).$$

Then

$$\sigma(\mathbf{y}) - \sigma(\mathbf{s}) = c\mathbf{v}.$$

Since $\sigma(\mathbf{x}), \sigma(\mathbf{s}) \in \mathcal{W}$, we have

$$\mathbf{0}, \mathbf{v} \in \mathcal{W} - \sigma(\mathbf{s}).$$

By hypothesis,

$$-(\mathcal{W} - \sigma(\mathbf{s})) \subset \left(1 + \frac{1}{\lambda} \right) (\mathcal{W} - \sigma(\mathbf{s})).$$

Hence

$$-\frac{\mathbf{v}}{1 + \lambda^{-1}} \in \mathcal{W} - \sigma(\mathbf{s}).$$

Let

$$t = -(1 + \lambda^{-1})c \in (0, 1].$$

Since $\mathcal{W} - \sigma(\mathbf{s})$ is convex and contains $\mathbf{0}$, we get

$$c\mathbf{v} = t \left(-\frac{\mathbf{v}}{1 + \lambda^{-1}} \right) + (1 - t)\mathbf{0} \in \mathcal{W} - \sigma(\mathbf{s}).$$

Therefore $\sigma(\mathbf{y}) \in \mathcal{W}$, a contradiction to the visibility of $\mathbf{x}$ from $\mathbf{s}$. This proves

$$\Lambda_{\text{vis}, \{\mathbf{s}\}} \subseteq \left\{ \mathbf{x} \in \Lambda : \gcd(\mathbf{x} - \mathbf{s}) = 1, \sigma\left(\frac{\mathbf{x} - \mathbf{s}}{\lambda}\right) \notin \mathcal{W} - \sigma(\mathbf{s}) \right\}. \quad (9)$$

We now prove the reverse inclusion $\supseteq$ in (7). Suppose, toward a contradiction, that

$$\mathbf{x} \in \Lambda \setminus \Lambda_{\text{vis}, \{\mathbf{s}\}}, \quad \gcd(\mathbf{x} - \mathbf{s}) = 1, \quad \sigma\left(\frac{\mathbf{x} - \mathbf{s}}{\lambda}\right) \notin \mathcal{W} - \sigma(\mathbf{s}).$$

Then there exists a point $\mathbf{y} \in \Lambda$ that is also in the interior of the straight line segment between $\mathbf{s}$ and $\mathbf{x}$. Write $\mathbf{y} = \mathbf{s} + t(\mathbf{x} - \mathbf{s})$ with $t \in (0, 1) \cap K$. Since Λ is locally finite, we may choose t minimal such that $\mathbf{y} \in \Lambda$. Then $\mathbf{y} \in \Lambda_{\text{vis}, \{\mathbf{s}\}}$.

Since $t \in K$, we may write $t = b/a$ with $a, b \in \mathcal{O}_K$ coprime. If b is not a unit, then

$$\mathbf{y} - \mathbf{s} = \frac{b}{a}(\mathbf{x} - \mathbf{s})$$

implies that $b \mid (\mathbf{y} - \mathbf{s})$, contradicting $\gcd(\mathbf{y} - \mathbf{s}) = 1$, which follows from the already proved inclusion (9). Thus b is a unit, and we may write $a = t^{-1} \in \mathcal{O}_K$.

If a is not a unit, then $\gcd(\mathbf{x} - \mathbf{s}) \neq 1$, which is impossible. Suppose that a is a unit. Since $t = a^{-1} \in (0, 1)$, it follows that $a = \lambda^k$ for some $k \in \mathbb{N}$. Since $\mathbf{y} = \mathbf{s} + \lambda^{-k}(\mathbf{x} - \mathbf{s}) \in \Lambda$, we have

$$\sigma(\mathbf{y}) = \sigma(\mathbf{s}) + \sigma(\lambda^{-k})\sigma(\mathbf{x} - \mathbf{s}) \in \mathcal{W},$$

and hence

$$\sigma(\lambda^{-k})\sigma(\mathbf{x} - \mathbf{s}) \in \mathcal{W} - \sigma(\mathbf{s}). \quad (10)$$

Observe that since $\sigma(\lambda) = -1/\lambda$, we must have

$$|\sigma(\lambda^{-k})| = (-1)^k \sigma(\lambda^{-k}) = \lambda^k \geq -\sigma(\lambda^{-1}) = \lambda > 1.$$

Recall that $\mathcal{W} - \sigma(\mathbf{s})$ is convex and contains $\mathbf{0}$. We distinguish two cases according to the parity of k .

Case 1. k is odd.

If $k = 1$ then (10) implies

$$\sigma(\lambda^{-1})\sigma(\mathbf{x} - \mathbf{s}) \in \mathcal{W} - \sigma(\mathbf{s}).$$

If $k \geq 3$, letting $\ell = \sigma(\lambda^{k-1}) \in (0, 1]$, (10) implies that

$$\sigma(\lambda^{-1})\sigma(\mathbf{x} - \mathbf{s}) = \ell\sigma(\lambda^{-k})\sigma(\mathbf{x} - \mathbf{s}) \in \mathcal{W} - \sigma(\mathbf{s}).$$

Case 2. k is even.

Combining this with (10), we obtain

$$\sigma(\lambda^{-1})^2\sigma(\mathbf{x} - \mathbf{s}) = -\lambda\sigma(\lambda^{-1})\sigma(\mathbf{x} - \mathbf{s}) \in \mathcal{W} - \sigma(\mathbf{s}).$$

Equivalently,

$$\lambda\sigma(\lambda^{-1})\sigma(\mathbf{x} - \mathbf{s}) \in -(\mathcal{W} - \sigma(\mathbf{s})).$$

By the hypothesis

$$-(\mathcal{W} - \sigma(\mathbf{s})) \subset \left(1 + \frac{1}{\lambda}\right) (\mathcal{W} - \sigma(\mathbf{s})),$$

we get

$$\lambda\sigma(\lambda^{-1})\sigma(\mathbf{x} - \mathbf{s}) \in \left(1 + \frac{1}{\lambda}\right) (\mathcal{W} - \sigma(\mathbf{s})).$$

By Lemma 3.1, $1 + \frac{1}{\lambda} \leq \lambda$. Since $\mathcal{W} - \sigma(\mathbf{s})$ is convex and contains $\mathbf{0}$, the dilates are nested under positive scaling. Hence

$$\lambda\sigma(\lambda^{-1})\sigma(\mathbf{x} - \mathbf{s}) \in \lambda(\mathcal{W} - \sigma(\mathbf{s})),$$

and therefore

$$\sigma(\lambda^{-1})\sigma(\mathbf{x} - \mathbf{s}) \in \mathcal{W} - \sigma(\mathbf{s}).$$

In both cases,

$$\sigma\left(\frac{\mathbf{x} - \mathbf{s}}{\lambda}\right) \in \mathcal{W} - \sigma(\mathbf{s}),$$

contradicting our assumption. This proves the reverse inclusion and completes the proof. $\square$

4. PROOF OF THEOREM 1.1

In order to prove Theorem 1.1 we will establish a series of auxiliary lemmas. In Section 4.2 we deduce the proof.

4.1. Window exclusions.

Lemma 4.1. *Let $\mathcal{W} \subseteq \mathbb{R}^d$ be a bounded, convex, and centrally symmetric set, let $\mathbf{s} \in \Lambda(\mathcal{W}, \mathcal{L})$, and let $\lambda > 1$. Assume that*

$$-(\mathcal{W} - \sigma(\mathbf{s})) \subseteq \left(1 + \frac{1}{\lambda}\right) (\mathcal{W} - \sigma(\mathbf{s}))$$

and

$$1 + \frac{1}{\lambda} \leq \lambda.$$

Define

$$\mathcal{W}_s \stackrel{\text{def}}{=} \frac{1}{\lambda} (\mathcal{W} + (\lambda + 1)\sigma(\mathbf{s})).$$

Then

$$\mathcal{W}_s \subseteq \mathcal{W}.$$

Proof. Since $\mathbf{s} \in \Lambda(\mathcal{W}, \mathcal{L})$, we have $\sigma(\mathbf{s}) \in \mathcal{W}$. Hence

$$\mathbf{0} \in \mathcal{W} - \sigma(\mathbf{s}).$$

Since $\mathcal{W}$ is convex, the translated set $\mathcal{W} - \sigma(\mathbf{s})$ is also convex. Therefore, because it contains $\mathbf{0}$, we have

$$t(\mathcal{W} - \sigma(\mathbf{s})) \subseteq \mathcal{W} - \sigma(\mathbf{s}) \text{ for every } t \in [0, 1].$$

Let $\mathbf{y} \in \mathcal{W}_s$. By definition, there exists $\mathbf{w} \in \mathcal{W}$ such that

$$\mathbf{y} = \frac{\mathbf{w} + (\lambda + 1)\sigma(\mathbf{s})}{\lambda} = \sigma(\mathbf{s}) + \frac{\mathbf{w} + \sigma(\mathbf{s})}{\lambda}.$$

Thus

$$\mathbf{y} - \sigma(\mathbf{s}) = \frac{\mathbf{w} + \sigma(\mathbf{s})}{\lambda}.$$

Since $\mathcal{W}$ is centrally symmetric, $-\mathbf{w} \in \mathcal{W}$. Hence

$$\mathbf{y} - \sigma(\mathbf{s}) = -\frac{1}{\lambda} ((-\mathbf{w}) - \sigma(\mathbf{s})) \in -\frac{1}{\lambda} (\mathcal{W} - \sigma(\mathbf{s})).$$

By hypothesis,

$$-(\mathcal{W} - \sigma(\mathbf{s})) \subseteq \left(1 + \frac{1}{\lambda}\right) (\mathcal{W} - \sigma(\mathbf{s})).$$

Multiplying by $1/\lambda$, we obtain

$$-\frac{1}{\lambda} (\mathcal{W} - \sigma(\mathbf{s})) \subseteq \frac{1 + \lambda^{-1}}{\lambda} (\mathcal{W} - \sigma(\mathbf{s})).$$

Since

$$0 < \frac{1 + \lambda^{-1}}{\lambda} \leq 1,$$

and since $\mathcal{W} - \sigma(\mathbf{s})$ is convex and contains $\mathbf{0}$, it follows that

$$\frac{1 + \lambda^{-1}}{\lambda} (\mathcal{W} - \sigma(\mathbf{s})) \subseteq \mathcal{W} - \sigma(\mathbf{s}).$$

Therefore

$$\mathbf{y} - \sigma(\mathbf{s}) \in \mathcal{W} - \sigma(\mathbf{s}),$$

and hence $\mathbf{y} \in \mathcal{W}$. Since $\mathbf{y} \in \mathcal{W}_s$ was arbitrary, we conclude that

$$\mathcal{W}_s \subseteq \mathcal{W}.$$

□

We denote the set of $\mathbf{s}$ -primitive points in $\Lambda(\mathcal{W}_s, \mathcal{L})$ by

$$\Lambda_{\text{pr},\{\mathbf{s}\}}(\mathcal{W}_s, \mathcal{L}) \stackrel{\text{def}}{=} \{\mathbf{x} \in \Lambda(\mathcal{W}_s, \mathcal{L}) : \gcd(\mathbf{x} - \mathbf{s}) = 1\}.$$

More generally, if $\mathcal{W}' \subset \mathcal{W}$, we denote the set of S -primitive points in $\Lambda(\mathcal{W}', \mathcal{L})$ by

$$\Lambda_{\text{pr},S}(\mathcal{W}', \mathcal{L}) \stackrel{\text{def}}{=} \{\mathbf{x} \in \Lambda(\mathcal{W}', \mathcal{L}) : \gcd(\mathbf{x} - \mathbf{s}) = 1 \text{ for all } \mathbf{s} \in S\}.$$

Let

$$A \stackrel{\text{def}}{=} \Lambda_{\text{pr},S}(\mathcal{W}, \mathcal{L}), \quad B_s \stackrel{\text{def}}{=} \Lambda_{\text{pr},\{\mathbf{s}\}}(\mathcal{W}_s, \mathcal{L}).$$

By Lemmas 2.2 and 4.1, we may write

$$\begin{aligned} \Lambda_{\text{vis},S} &= \Lambda_{\text{pr},S}(\mathcal{W}, \mathcal{L}) \setminus \bigcup_{\mathbf{s} \in S} (\Lambda_{\text{pr},\{\mathbf{s}\}}(\mathcal{W}_s, \mathcal{L}) \cap \Lambda_{\text{pr},S}(\mathcal{W}, \mathcal{L})) \\ &= A \setminus \bigcup_{\mathbf{s} \in S} (A \cap B_s). \end{aligned} \tag{11}$$

Applying inclusion–exclusion to the union in (11), we obtain

$$\begin{aligned} \#(\Lambda_{\text{vis},S} \cap TD) &= \#(A \cap TD) \\ &\quad - \sum_{\emptyset \neq I \subseteq S} (-1)^{\#(I)+1} \# \left(\left(\bigcap_{\mathbf{s} \in I} (A \cap B_s) \right) \cap TD \right). \end{aligned}$$

Since A is independent of $\mathbf{s}$, for every nonempty $I \subseteq S$ we have

$$\bigcap_{\mathbf{s} \in I} (A \cap B_s) = A \cap \bigcap_{\mathbf{s} \in I} B_s.$$

Substituting the definitions of A and B_s gives

$$\begin{aligned} \#(\Lambda_{\text{vis},S} \cap TD) &= \#(\Lambda_{\text{pr},S}(\mathcal{W}, \mathcal{L}) \cap TD) \\ &\quad - \sum_{\emptyset \neq I \subseteq S} (-1)^{\#(I)+1} \# \left(\Lambda_{\text{pr},S}(\mathcal{W}, \mathcal{L}) \cap \bigcap_{\mathbf{s} \in I} \Lambda_{\text{pr},\{\mathbf{s}\}}(\mathcal{W}_s, \mathcal{L}) \cap TD \right). \end{aligned} \tag{12}$$

Lemma 4.2. *Let $\mathcal{W} \subseteq \mathbb{R}^d$ be a bounded, convex, and centrally symmetric set, let $\mathbf{s} \in \Lambda(\mathcal{W}, \mathcal{L})$, and let $\lambda > 1$. Assume that*

$$-(\mathcal{W} - \sigma(\mathbf{s})) \subseteq \left(1 + \frac{1}{\lambda}\right) (\mathcal{W} - \sigma(\mathbf{s}))$$

and

$$1 + \frac{1}{\lambda} \leq \lambda.$$

Let $I \subseteq S$ be nonempty. Then

$$\Lambda_{\text{pr},S}(\mathcal{W}, \mathcal{L}) \cap \bigcap_{\mathbf{s} \in I} \Lambda_{\text{pr},\{\mathbf{s}\}}(\mathcal{W}_s, \mathcal{L}) = \Lambda_{\text{pr},S}(\mathcal{W}_I, \mathcal{L}).$$

Proof. Let $\mathbf{x} \in \Lambda_{\text{pr},S}(\mathcal{W}, \mathcal{L}) \cap \bigcap_{\mathbf{s} \in I} \Lambda_{\text{pr},\{\mathbf{s}\}}(\mathcal{W}_s, \mathcal{L})$. Since $\mathbf{x} \in \Lambda_{\text{pr},S}(\mathcal{W}, \mathcal{L})$, we have

$$\gcd(\mathbf{x} - \mathbf{s}) = 1 \quad \text{for all } \mathbf{s} \in S.$$

Moreover, for every $\mathbf{s} \in I$, the condition $\mathbf{x} \in \Lambda_{\text{pr},\{\mathbf{s}\}}(\mathcal{W}_s, \mathcal{L})$ implies $\mathbf{x} \in \Lambda(\mathcal{W}_s, \mathcal{L})$. Hence

$$\sigma(\mathbf{x}) \in \bigcap_{\mathbf{s} \in I} \mathcal{W}_s = \mathcal{W}_I,$$

and therefore $\mathbf{x} \in \Lambda(\mathcal{W}_I, \mathcal{L})$. Combining this with the coprimality conditions for all $\mathbf{s} \in S$, we obtain

$$\mathbf{x} \in \Lambda_{\text{pr}, S}(\mathcal{W}_I, \mathcal{L}).$$

Thus

$$\Lambda_{\text{pr}, S}(\mathcal{W}, \mathcal{L}) \cap \bigcap_{\mathbf{s} \in I} \Lambda_{\text{pr}, \{\mathbf{s}\}}(\mathcal{W}_{\mathbf{s}}, \mathcal{L}) \subseteq \Lambda_{\text{pr}, S}(\mathcal{W}_I, \mathcal{L}).$$

Conversely, let $\mathbf{x} \in \Lambda_{\text{pr}, S}(\mathcal{W}_I, \mathcal{L})$. Then

$$\sigma(\mathbf{x}) \in \mathcal{W}_I = \bigcap_{\mathbf{s} \in I} \mathcal{W}_{\mathbf{s}}$$

and

$$\gcd(\mathbf{x} - \mathbf{s}) = 1 \quad \text{for all } \mathbf{s} \in S.$$

Since $\mathcal{W}_I \subseteq \mathcal{W}_{\mathbf{s}}$ for every $\mathbf{s} \in I$, it follows that

$$\mathbf{x} \in \Lambda_{\text{pr}, \{\mathbf{s}\}}(\mathcal{W}_{\mathbf{s}}, \mathcal{L}) \quad \text{for every } \mathbf{s} \in I.$$

Also, by Lemma 4.1, we have $\mathcal{W}_{\mathbf{s}} \subseteq \mathcal{W}$ for every $\mathbf{s} \in I$, and hence

$$\mathcal{W}_I \subseteq \mathcal{W}.$$

Therefore $\mathbf{x} \in \Lambda(\mathcal{W}, \mathcal{L})$. Together with the coprimality conditions for all $\mathbf{s} \in S$, this gives

$$\mathbf{x} \in \Lambda_{\text{pr}, S}(\mathcal{W}, \mathcal{L}).$$

Thus

$$\Lambda_{\text{pr}, S}(\mathcal{W}_I, \mathcal{L}) \subseteq \Lambda_{\text{pr}, S}(\mathcal{W}, \mathcal{L}) \cap \bigcap_{\mathbf{s} \in I} \Lambda_{\text{pr}, \{\mathbf{s}\}}(\mathcal{W}_{\mathbf{s}}, \mathcal{L}),$$

and the proof is complete. $\square$

Using (12) and Lemma 4.2, we obtain the following corollary.

Corollary 4.3. *Under the hypotheses of Theorem 1.1, for every sufficiently large $T > 0$ we have that*

$$\begin{aligned} \#(\Lambda_{\text{vis}, S}(\mathcal{W}, \mathcal{L}) \cap TD) &= \#(\Lambda_{\text{pr}, S}(\mathcal{W}, \mathcal{L}) \cap TD) \\ &\quad - \sum_{\emptyset \neq I \subseteq S} (-1)^{|I|+1} \#(\Lambda_{\text{pr}, S}(\mathcal{W}_I, \mathcal{L}) \cap TD). \end{aligned}$$

The final step in the proof of Theorem 1.1 is the following lemma.

Lemma 4.4. *Under the hypotheses of Theorem 1.1, let either $\mathcal{W}' = \mathcal{W}$, or $\mathcal{W}' = \mathcal{W}_I$ for some nonempty subset $I \subseteq S$. Then the density of $\Lambda_{\text{pr}, S}(\mathcal{W}', \mathcal{L})$ along $\{TD\}_{T>0}$ exists and is given by:*

$$\theta(\Lambda_{\text{pr}, S}(\mathcal{W}', \mathcal{L})) = \frac{\text{vol}(\mathcal{W}')}{\text{covol}(\mathcal{L})} \prod_{\mathfrak{p} \in \mathcal{P}_{\mathcal{O}_K}} \left(1 - \frac{s(\mathfrak{p})}{N(\mathfrak{p})^d}\right).$$

4.2. Completion of the proof of Theorem 1.1. Assuming Lemma 4.4, we now complete the proof of Theorem 1.1. By Corollary 4.3, the density of $\Lambda_{\text{vis},S}(\mathcal{W}, \mathcal{L})$ along the sequence $\{TD\}_{T>0}$ exists. Moreover,

$$\begin{aligned} \theta(\Lambda_{\text{vis},S}(\mathcal{W}, \mathcal{L})) &= \lim_{T \rightarrow \infty} \frac{\#(\Lambda_{\text{pr},S}(\mathcal{W}, \mathcal{L}) \cap TD)}{\text{vol}(TD)} \\ &\quad - \sum_{\emptyset \neq I \subseteq S} (-1)^{\#(I)+1} \lim_{T \rightarrow \infty} \frac{\#(\Lambda_{\text{pr},S}(\mathcal{W}_I, \mathcal{L}) \cap TD)}{\text{vol}(TD)} \\ &= \frac{\text{vol}(\mathcal{W})}{\text{covol}(\mathcal{L})} \prod_{\mathfrak{p} \in \mathcal{P}_{\mathcal{O}_K}} \left(1 - \frac{s(\mathfrak{p})}{N(\mathfrak{p})^d}\right) \\ &\quad - \sum_{\emptyset \neq I \subseteq S} (-1)^{\#(I)+1} \frac{\text{vol}(\mathcal{W}_I)}{\text{covol}(\mathcal{L})} \prod_{\mathfrak{p} \in \mathcal{P}_{\mathcal{O}_K}} \left(1 - \frac{s(\mathfrak{p})}{N(\mathfrak{p})^d}\right) \\ &= \text{dens}(\mathcal{L}) \left(\text{vol}(\mathcal{W}) - \sum_{\emptyset \neq I \subseteq S} (-1)^{\#(I)+1} \text{vol}(\mathcal{W}_I) \right) \prod_{\mathfrak{p} \in \mathcal{P}_{\mathcal{O}_K}} \left(1 - \frac{s(\mathfrak{p})}{N(\mathfrak{p})^d}\right). \end{aligned}$$

This is the formula claimed in Theorem 1.1.

It remains to prove Lemma 4.4.

4.3. Proof of Lemma 4.4. For a rational prime p , let M_p denote the set of prime ideals of $\mathcal{O}_K$ lying above p . Since K is quadratic, we have

$$\#(M_p) \leq 2.$$

Let p_i denote the i th rational prime, and define

$$\mathcal{E}_t \stackrel{\text{def}}{=} \bigcup_{i=1}^t M_{p_i}, \quad t \in \mathbb{N}.$$

For a prime ideal $\mathfrak{p}$, define

$$\mathfrak{p}^d \stackrel{\text{def}}{=} \{(x_1, \dots, x_m) \in \mathcal{O}_K^d : x_i \in \mathfrak{p}\}.$$

For $\mathcal{W}'$ as in Lemma 4.4, define

$$\Lambda_{\text{pr},S,\mathcal{E}_t}(\mathcal{W}', \mathcal{L}) \stackrel{\text{def}}{=} \{\mathbf{x} \in \Lambda(\mathcal{W}', \mathcal{L}) : \mathbf{x} - \mathbf{s} \notin \mathfrak{p}^d \text{ for all } \mathfrak{p} \in \mathcal{E}_t \text{ and all } \mathbf{s} \in S\}.$$

Equivalently, $\Lambda_{\text{pr},S,\mathcal{E}_t}(\mathcal{W}', \mathcal{L})$ is obtained from $\Lambda(\mathcal{W}', \mathcal{L})$ by imposing only the primitivity conditions coming from prime ideals above the first t rational primes.

In the following lemma, we show that the density of $\Lambda_{\text{pr},S,\mathcal{E}_t}(\mathcal{W}', \mathcal{L})$ along averaging sets TD exists.

Lemma 4.5. *Under the hypotheses of Lemma 4.4, the density of $\Lambda_{\text{pr},S,\mathcal{E}_t}(\mathcal{W}', \mathcal{L})$ along the sequence $\{TD\}_{T>0}$ exists. Moreover,*

$$\theta(\Lambda_{\text{pr},S,\mathcal{E}_t}) = \frac{\text{vol}(\mathcal{W}')}{\text{covol}(\mathcal{L})} \cdot \prod_{\mathfrak{p} \in \mathcal{E}_t} \left(1 - \frac{s(\mathfrak{p})}{N(\mathfrak{p})^d}\right).$$

Proof. Let $G = \mathbb{R}^d$ be the physical space, and let

$$H = \mathbb{R}^d \times \prod_{\mathfrak{p} \in \mathcal{E}_t} (\mathcal{O}_K/\mathfrak{p})^d$$

be the internal space. Both G and H are σ -compact and LCAGs, and the group H is compactly generated. Let $\mathcal{L}'$ be the image of the diagonal embedding

$$\mathcal{O}_K^d \hookrightarrow G \times H \quad \mathbf{x} \mapsto (\mathbf{x}, \sigma(\mathbf{x}), (\pi_{\mathfrak{p}}(\mathbf{x}))_{\mathfrak{p} \in \mathcal{E}_t}).$$

Then $\mathcal{L}'$ is a lattice. The projection of $\mathcal{L}'$ to G is injective, while the projection to H is dense. Moreover, $\text{covol}(\mathcal{L}') = \text{covol}(\mathcal{L})$.

Define

$$\mathcal{W}'' \stackrel{\text{def}}{=} \mathcal{W}' \times \prod_{\mathfrak{p} \in \mathcal{E}_t} \mathcal{W}_{\mathfrak{p}},$$

where

$$\mathcal{W}_{\mathfrak{p}} \stackrel{\text{def}}{=} \left(\mathcal{O}_K / \mathfrak{p} \right)^d \setminus \pi_{\mathfrak{p}}(S).$$

The set $\mathcal{W}''$ is a regular window: it is relatively compact, has nonempty interior, and its boundary has Haar measure zero. By construction, the associated cut-and-project set $\Lambda(\mathcal{W}'', \mathcal{L}')$ consists exactly of those $\mathbf{x} \in \mathcal{O}_K^d$ such that

$$\sigma(\mathbf{x}) \in \mathcal{W}' \quad \text{and} \quad \pi_{\mathfrak{p}}(\mathbf{x}) \notin \pi_{\mathfrak{p}}(S) \quad \text{for every } \mathfrak{p} \in \mathcal{E}_t.$$

Equivalently,

$$\Lambda(\mathcal{W}'', \mathcal{L}') = \Lambda_{\text{pr}, S, \mathcal{E}_t}(\mathcal{W}', \mathcal{L}).$$

Furthermore,

$$\text{vol}(\mathcal{W}'') = \text{vol}(\mathcal{W}') \prod_{\mathfrak{p} \in \mathcal{E}_t} \left(1 - \frac{s(\mathfrak{p})}{N(\mathfrak{p})^d} \right),$$

where the finite factors are equipped with normalized counting measure. It follows from the construction that $\Lambda(\mathcal{W}'', \mathcal{L}')$ is an irreducible cut-and-project set. Applying the density formula (6) for irreducible cut-and-project sets, we get

$$\begin{aligned} \theta(\Lambda_{\text{pr}, S, \mathcal{E}_t}) &= \theta(\Lambda(\mathcal{W}'', \mathcal{L}')) \\ &= \frac{\text{vol}(\mathcal{W}')}{\text{covol}(\mathcal{L}')} \prod_{\mathfrak{p} \in \mathcal{E}_t} \left(1 - \frac{s(\mathfrak{p})}{N(\mathfrak{p})^d} \right) \\ &= \frac{\text{vol}(\mathcal{W}')}{\text{covol}(\mathcal{L})} \prod_{\mathfrak{p} \in \mathcal{E}_t} \left(1 - \frac{s(\mathfrak{p})}{N(\mathfrak{p})^d} \right). \end{aligned}$$

□

Since

$$\Lambda_{\text{pr}, S}(\mathcal{W}', \mathcal{L}) \subseteq \Lambda_{\text{pr}, S, \mathcal{E}_t}(\mathcal{W}', \mathcal{L})$$

for every $t \in \mathbb{N}$, Lemma 4.5 gives

$$\bar{\theta}(\Lambda_{\text{pr}, S}(\mathcal{W}', \mathcal{L})) \leq \theta(\Lambda_{\text{pr}, S, \mathcal{E}_t}(\mathcal{W}', \mathcal{L})) = \frac{\text{vol}(\mathcal{W}')}{\text{covol}(\mathcal{L})} \prod_{\mathfrak{p} \in \mathcal{E}_t} \left(1 - \frac{s(\mathfrak{p})}{N(\mathfrak{p})^d} \right).$$

Letting $t \rightarrow \infty$, we obtain

$$\bar{\theta}(\Lambda_{\text{pr}, S}(\mathcal{W}', \mathcal{L})) \leq \frac{\text{vol}(\mathcal{W}')}{\text{covol}(\mathcal{L})} \prod_{\mathfrak{p} \in \mathcal{P}_{\mathcal{O}_K}} \left(1 - \frac{s(\mathfrak{p})}{N(\mathfrak{p})^d} \right).$$

Lemma 4.6. *For every $t \in \mathbb{N}$,*

$$\underline{\theta}(\Lambda_{\text{pr}, S}(\mathcal{W}', \mathcal{L})) = \theta(\Lambda_{\text{pr}, S, \mathcal{E}_t}(\mathcal{W}', \mathcal{L})) - \bar{\theta}(\Lambda_{\text{pr}, S, \mathcal{E}_t}(\mathcal{W}', \mathcal{L}) \setminus \Lambda_{\text{pr}, S}(\mathcal{W}', \mathcal{L})).$$

Proof. Let $A \stackrel{\text{def}}{=} \Lambda_{\text{pr}, S, \mathcal{E}_t}(\mathcal{W}', \mathcal{L})$ and $B \stackrel{\text{def}}{=} \Lambda_{\text{pr}, S}(\mathcal{W}', \mathcal{L})$. Because $B \subseteq A$, we can decompose A into the disjoint union

$$A = B \sqcup (A \setminus B).$$

Intersecting both sides with TD gives the finite counting identity for every $T > 0$

$$\#(B \cap TD) = \#(A \cap TD) - \#((A \setminus B) \cap TD).$$

Dividing all terms by $\text{vol}(TD)$ yields:

$$\frac{\#(B \cap TD)}{\text{vol}(TD)} = \frac{\#(A \cap TD)}{\text{vol}(TD)} - \frac{\#((A \setminus B) \cap TD)}{\text{vol}(TD)}.$$

Let

$$a_T \stackrel{\text{def}}{=} \frac{\#(A \cap TD)}{\text{vol}(TD)} \quad \text{and} \quad c_T \stackrel{\text{def}}{=} \frac{\#((A \setminus B) \cap TD)}{\text{vol}(TD)}.$$

Let T_n be a sequence such that $T_n \rightarrow \infty$ and

$$\lim_{n \rightarrow \infty} c_{T_n} = \limsup_{T \rightarrow \infty} c_T.$$

By Lemma 4.5, the sequence a_T converges to the density $\theta(A)$ as $T \rightarrow \infty$. Hence $\lim_{n \rightarrow \infty} a_{T_n} = \theta(A)$. On one hand

$$a - \limsup_{T \rightarrow \infty} c_T = \lim_{n \rightarrow \infty} a_{T_n} - \lim_{n \rightarrow \infty} c_{T_n} = \lim_{n \rightarrow \infty} (a_{T_n} - c_{T_n}) \geq \liminf_{T \rightarrow \infty} (a_T - c_T).$$

On the other

$$\liminf_{T \rightarrow \infty} (a_T - c_T) \geq \liminf_{T \rightarrow \infty} a_T + \liminf_{T \rightarrow \infty} (-c_T) = a - \limsup_{T \rightarrow \infty} c_T.$$

We obtain:

$$\underline{\theta}(B) = \theta(A) - \bar{\theta}(A \setminus B).$$

□

The proof of Lemma 4.4 will follow from the following lemma in combination with Lemma 4.6.

Lemma 4.7. *Under the hypotheses of Lemma 4.4,*

$$\lim_{t \rightarrow \infty} \bar{\theta}(\Lambda_{\text{pr}, S, \mathcal{E}_t}(\mathcal{W}', \mathcal{L}) \setminus \Lambda_{\text{pr}, S}(\mathcal{W}', \mathcal{L})) = 0.$$

Proof. Define

$$A_t \stackrel{\text{def}}{=} \Lambda_{\text{pr}, S, \mathcal{E}_t}(\mathcal{W}', \mathcal{L}) \setminus \Lambda_{\text{pr}, S}(\mathcal{W}', \mathcal{L}).$$

Thus, A_t consists of those points that satisfy the primitivity conditions for every $\mathbf{s} \in S$ and every prime ideal $\mathbf{p} \in \mathcal{E}_t$. However, they fail the primitivity condition for at least one pair $(s, \mathbf{p})$, where $\mathbf{s} \in S$ and $\mathbf{p} \notin \mathcal{E}_t$.

For $g \in \mathcal{O}_K$, let

$$a_g \stackrel{\text{def}}{=} \text{diag}(\underbrace{g, \dots, g}_{d \text{ times}}, \underbrace{\sigma(g), \dots, \sigma(g)}_{d \text{ times}}),$$

and define

$$\mathcal{L}_{(g)} \stackrel{\text{def}}{=} a_g \mathcal{L} = \{(g\mathbf{x}, \sigma(g\mathbf{x})) : \mathbf{x} \in \mathcal{O}_K^d\}.$$

Since $\mathcal{O}_K$ is a ring, $\mathcal{L}_{(g)} \subseteq \mathcal{L}$, and

$$\text{covol}(\mathcal{L}_{(g)}) = |N(g)|^d \text{covol}(\mathcal{L}).$$

As $\mathcal{O}_K$ is a principal ideal domain, if $\mathfrak{p} = (g)$ we write $\mathcal{L}_{\mathfrak{p}}$ for $\mathcal{L}_{(g)}$. We claim that

$$A_t \subseteq \bigcup_{s \in S} \bigcup_{\substack{\mathfrak{p} \in \mathcal{P}_{\mathcal{O}_K} \\ \mathfrak{p} \notin \mathcal{E}_t}} (s + \Lambda(\mathcal{W}' - \sigma(s), \mathcal{L}_{\mathfrak{p}})). \quad (13)$$

Indeed, if $\mathbf{x} \in A_t$, there exist $\mathbf{s}_0 \in S$ and a prime ideal $\mathfrak{p} \notin \mathcal{E}_t$ such that

$$\mathfrak{p} \mid (\mathbf{x} - \mathbf{s}_0),$$

or equivalently $\mathbf{x} - \mathbf{s}_0 \in \mathfrak{p}^d$. Since also $\sigma(\mathbf{x}) \in \mathcal{W}'$, we have

$$\sigma(\mathbf{x} - \mathbf{s}_0) \in \mathcal{W}' - \sigma(\mathbf{s}_0).$$

Thus

$$\mathbf{x} \in \mathbf{s}_0 + \Lambda(\mathcal{W}' - \sigma(\mathbf{s}_0), \mathcal{L}_{\mathfrak{p}}),$$

which proves (13).

Claim. *There exists a positive constant $c > 0$ such that for all $t \in \mathbb{N}$ and for all large T , we have*

$$A_t \cap TD \subseteq \bigcup_{s \in S} \bigcup_{\substack{\mathfrak{p} \in \mathcal{P}_{\mathcal{O}_K} \\ \mathfrak{p} \notin \mathcal{E}_t \\ N(\mathfrak{p}) \leq cT}} (s + \Lambda(\mathcal{W}' - \sigma(s), \mathcal{L}_{\mathfrak{p}}) \cap TD). \quad (14)$$

That is, the union on the right-hand side of (14) is taken over finitely many prime ideals $\mathfrak{p} \notin \mathcal{E}_t$ with norm at most cT .

Proof of claim. Fix $\mathbf{s} \in S$, and suppose

$$\mathbf{x} \in (s + \Lambda(\mathcal{W}' - \sigma(s), \mathcal{L}_{\mathfrak{p}})) \cap TD, \quad \mathbf{x} \neq \mathbf{s}.$$

Then $\mathbf{x} - \mathbf{s} \in \mathfrak{p}^d$. Choose a coordinate j such that $x_j - s_j \neq 0$. Because $x_j - s_j \in \mathfrak{p}$, we have $(x_j - s_j) \subseteq \mathfrak{p}$, where $(x_j - s_j)$ is the ideal generated by $x_j - s_j$. For ideals, inclusion reverses the norm inequality; hence

$$N(\mathfrak{p}) \leq N((x_j - s_j)) = |N(x_j - s_j)|.$$

Now $N(x_j - s_j) = (x_j - s_j)\sigma(x_j - s_j)$. Since D is bounded and $\mathbf{x} \in TD$, there is a constant $C_D > 0$ depending only on D , such that $|x_j| \leq C_D T$. Since S is finite, we obtain

$$|x_j - s_j| \leq c_1 T, \quad \text{for all } 1 \leq j \leq d, \text{ and all } \mathbf{s} \in S, \quad (15)$$

where c_1 is a constant independent of $\mathfrak{p}, \mathbf{s}, T$. Also, $\sigma(x_j - s_j) = \sigma(x_j) - \sigma(s_j)$. The window $\mathcal{W}'$ is bounded. Thus, given $\mathbf{y} = (y_1, \dots, y_d) \in \mathcal{W} - \sigma(\mathbf{s})$, there exists $c_s > 0$ such that

$$|y_j| \leq c_s \quad \text{for all } 1 \leq j \leq d.$$

The constant $c_2 \stackrel{\text{def}}{=} \max_{\mathbf{s} \in S} \{c_s\} > 0$ is independent of $\mathbf{x}, \mathbf{s}, T$ and $\mathfrak{p}$, and satisfies

$$|\sigma(x_j - s_j)| \leq c_2. \quad (16)$$

Combining (15) and (16), we obtain

$$N(\mathfrak{p}) \leq c_1 c_2 T.$$

Let H_n be the number of ideals in $\mathcal{O}_K$ of norm n . It follows from [Mar18, Theorem 39] that

$$H_n \leq H \cdot n^{1/2}, \quad (17)$$

where $H > 0$ is a constant independent of n . Hence, for a fixed T , only finitely many prime ideals can occur in (14). This completes the proof of the claim. $\triangle$

For $\mathbf{s} \in S$ and $\mathbf{p} \in \mathcal{P}_{\mathcal{O}_K}$, write

$$N_{\mathbf{s},\mathbf{p}}(T) \stackrel{\text{def}}{=} \#((\mathbf{s} + \Lambda(\mathcal{W}' - \sigma(\mathbf{s}), \mathcal{L}_{\mathbf{p}})) \cap TD).$$

By unraveling the definitions,

$$\mathbf{s} + \Lambda(\mathcal{W}' - \sigma(\mathbf{s}), \mathcal{L}_{\mathbf{p}}) = \Lambda(\mathcal{W}', \mathcal{L}_{\mathbf{p}} + (\mathbf{s}, \sigma(\mathbf{s}))),$$

and

$$N_{\mathbf{s},\mathbf{p}}(T) = \#((TD - \mathbf{s}) \times (\mathcal{W}' - \sigma(\mathbf{s})) \cap \mathcal{L}_{\mathbf{p}}).$$

Hence, the sequence $N_{\mathbf{s},\mathbf{p}}(T)$ can be understood by counting lattice points in a convex set.

By (14), we have

$$\#(A_t \cap TD) \leq \sum_{\mathbf{s} \in S} \sum_{\mathbf{p} \notin \mathcal{E}_t} N_{\mathbf{s},\mathbf{p}}(T), \quad (18)$$

where the second sum runs over finitely many primes. By [GKW25, Lemma 5.4] we have for $N(\mathbf{p}) \leq T$:

$$\begin{aligned} N_{\mathbf{s},\mathbf{p}}(T) &\leq \frac{\text{vol}(TD) \cdot \text{vol}(\mathcal{W}')}{\text{covol}(\mathcal{L}_{\mathbf{p}})} + C_1 \left(\frac{T}{N(\mathbf{p})} \right)^{d-\frac{1}{2}} \\ N_{\mathbf{s},\mathbf{p}}(T) &\leq C_0 \frac{T^d}{N(\mathbf{p})^d} + C_1 \left(\frac{T}{N(\mathbf{p})} \right)^{d-\frac{1}{2}} \\ \frac{N_{\mathbf{s},\mathbf{p}}(T)}{\text{vol}(TD)} &\leq \frac{C_2}{N(\mathbf{p})^d} + \frac{C_3}{T^{1/2} N(\mathbf{p})^{d-\frac{1}{2}}}, \end{aligned} \quad (19)$$

where the constants $C_0, C_1, C_2, C_3 > 0$, independent of $\mathbf{s}, \mathbf{p}$. It follows from [GKW25, Lemma 5.5], that for prime ideals satisfying $N(\mathbf{p}) > T$

$$\frac{N_{\mathbf{s},\mathbf{p}}(T)}{\text{vol}(TD)} \leq \frac{C_4}{N(\mathbf{p})^d}, \quad (20)$$

for some constant C_4 independent of $\mathbf{s}, \mathbf{p}$. Dividing $N_{\mathbf{s},\mathbf{p}}(T)$ over the sum of primes with $N(\mathbf{p}) \leq T$ and $N(\mathbf{p}) > T$, we obtain

$$N_{\mathbf{s},\mathbf{p}}(T) = \sum_{\substack{\mathbf{p} \notin \mathcal{E}_t \\ N(\mathbf{p}) \leq T}} N_{\mathbf{s},\mathbf{p}}(T) + \sum_{\substack{\mathbf{p} \notin \mathcal{E}_t \\ T < N(\mathbf{p}) \leq c_1 c_2 T}} N_{\mathbf{s},\mathbf{p}}(T)$$

By (18), (19) and (20), we obtain for some constant C independent of $\mathbf{s}$ and $\mathbf{p}$

$$\frac{\#(A_t \cap TD)}{\text{vol}(TD)} \leq C \cdot \#(S) \sum_{\substack{\mathbf{p} \notin \mathcal{E}_t \\ N(\mathbf{p}) \leq T}} \frac{1}{N(\mathbf{p})^d} + \frac{C \cdot \#(S)}{T^{1/2}} \sum_{\substack{\mathbf{p} \notin \mathcal{E}_t \\ N(\mathbf{p}) \leq T}} \frac{1}{N(\mathbf{p})^{d-\frac{1}{2}}} + C \cdot \#(S) \sum_{\substack{\mathbf{p} \notin \mathcal{E}_t \\ N(\mathbf{p}) > T}} \frac{1}{N(\mathbf{p})^d}.$$

Combining the first and third sums yields:

$$\frac{\#(A_t \cap TD)}{\text{vol}(TD)} \leq C \cdot \#(S) \sum_{\substack{\mathbf{p} \in \mathcal{P}_{\mathcal{O}_K} \\ \mathbf{p} \notin \mathcal{E}_t}} \frac{1}{N(\mathbf{p})^d} + \frac{C \cdot \#(S)}{T^{1/2}} \sum_{\mathbf{p} \in \mathcal{P}_{\mathcal{O}_K}} \frac{1}{N(\mathbf{p})^{d-\frac{1}{2}}}. \quad (21)$$

Because $d \geq 2$, we have $d - \frac{1}{2} > 1$, and hence,

$$\sum_{\mathbf{p} \in \mathcal{P}_{\mathcal{O}_K}} \frac{1}{N(\mathbf{p})^{d-\frac{1}{2}}} < \infty.$$

Taking $\limsup_{T \rightarrow \infty}$ in (21), we get

$$\bar{\theta}(A_t) \leq C \cdot \#(S) \sum_{\substack{\mathfrak{p} \in \mathcal{P}_{\mathcal{O}_K} \\ \mathfrak{p} \notin \mathcal{E}_t}} \frac{1}{N(\mathfrak{p})^d}. \quad (22)$$

Finally, since $d \geq 2$, $\sum_{\mathfrak{p} \in \mathcal{P}_{\mathcal{O}_K}} \frac{1}{N(\mathfrak{p})^d} < \infty$. The sets $\mathcal{E}_t$ exhaust $\mathcal{P}_{\mathcal{O}_K}$ as $t \rightarrow \infty$,

$$\sum_{\substack{\mathfrak{p} \in \mathcal{P}_{\mathcal{O}_K} \\ \mathfrak{p} \notin \mathcal{E}_t}} \frac{1}{N(\mathfrak{p})^d} \xrightarrow{t \rightarrow \infty} 0. \quad (23)$$

Combining (22) and (23), we conclude that

$$\lim_{t \rightarrow \infty} \bar{\theta}(A_t) = 0.$$

□

5. PROOF OF COROLLARY 1.2

Let $G = \mathbb{R}^d$ be the physical space, and let

$$H = \mathbb{R}^d \times \prod_{\mathfrak{p} \in \mathcal{E}_t} (\mathcal{O}_K/\mathfrak{p})^d$$

be the internal space as in the proof of Theorem 1.1. Let $\mathcal{L}_2$ be the image of the diagonal embedding

$$\mathcal{O}_K^d \hookrightarrow G \times H \quad \mathbf{x} \mapsto (\mathbf{x}, \sigma(\mathbf{x}), (\pi_{\mathfrak{p}}(\mathbf{x}))_{\mathfrak{p} \in \mathcal{P}_{\mathcal{O}_K}}).$$

Then $\mathcal{L}_2$ is a lattice. The projection of $\mathcal{L}_2$ to G is injective, while the projection to H is dense. Moreover, $\text{covol}(\mathcal{L}_2) = \text{covol}(\mathcal{L})$. Let

$$\mathcal{W}_2 = \mathcal{W} \setminus \left(\bigcup_{s \in S} \mathcal{W}_s \right) \times \prod_{\mathfrak{p} \in \mathcal{P}_{\mathcal{O}_K}} \mathcal{W}_{\mathfrak{p}},$$

where

$$\mathcal{W}_{\mathfrak{p}} \stackrel{\text{def}}{=} (\mathcal{O}_K/\mathfrak{p})^d \setminus \pi_{\mathfrak{p}}(S).$$

We claim that

$$\Lambda_{\text{vis}, S} = \Lambda(\mathcal{W}_2, \mathcal{L}_2).$$

In fact, by the definitions of $\Lambda_{\text{vis}, S}$, $\Lambda_{\text{vis}, \mathbf{s}}$, and Lemma 2.2, we have

$$\begin{aligned} \mathbf{x} \in \Lambda_{\text{vis}, S} &\iff \mathbf{x} \in \Lambda_{\text{vis}, \mathbf{s}}, \quad \forall \mathbf{s} \in S \\ &\iff \sigma(\mathbf{x}) \in \mathcal{W}, \sigma(\mathbf{x}) \notin \mathcal{W}_{\mathbf{s}}, \gcd(\mathbf{x} - \mathbf{s}) = 1, \forall \mathbf{s} \in S \\ &\iff \sigma(\mathbf{x}) \in \mathcal{W} \setminus \bigcup_{s \in S} \mathcal{W}_s, \gcd(\mathbf{x} - \mathbf{s}) = 1, \forall \mathbf{s} \in S \\ &\iff \sigma(\mathbf{x}) \in \mathcal{W} \setminus \bigcup_{s \in S} \mathcal{W}_s, \pi_{\mathfrak{p}}(\mathbf{x}) \notin \pi_{\mathfrak{p}}(\mathbf{x})(S) \\ &\iff \mathbf{x} \in \Lambda(\mathcal{W}_2, \mathcal{L}_2). \end{aligned}$$

Since $\mathcal{W}_s \subset \mathcal{W}$, inclusion–exclusion gives

$$\begin{aligned} \theta_H(\mathcal{W}_2) &= \text{vol} \left(\mathcal{W} \setminus \left(\bigcup_{s \in S} \mathcal{W}_s \right) \right) \cdot \prod_{\mathfrak{p} \in \mathcal{P}_{\mathcal{O}_K}} \left(1 - \frac{s(\mathfrak{p})}{N(\mathfrak{p})^d} \right) \\ &= \left(\text{vol}(\mathcal{W}) - \sum_{\emptyset \neq I \subseteq S} (-1)^{\#(I)+1} \text{vol}(\cap_{s \in I} \mathcal{W}_s) \right) \cdot \prod_{\mathfrak{p} \in \mathcal{P}_{\mathcal{O}_K}} \left(1 - \frac{s(\mathfrak{p})}{N(\mathfrak{p})^d} \right) \\ &= \left(\text{vol}(\mathcal{W}) - \sum_{\emptyset \neq I \subseteq S} (-1)^{\#(I)+1} \text{vol}(\mathcal{W}_I) \right) \cdot \prod_{\mathfrak{p} \in \mathcal{P}_{\mathcal{O}_K}} \left(1 - \frac{s(\mathfrak{p})}{N(\mathfrak{p})^d} \right). \end{aligned}$$

Since $\text{dens}(\mathcal{L}) = \text{dens}(\mathcal{L}_2)$, by Theorem 1.1, we get

$$\theta(\Lambda_{\text{vis},S}) = \theta(\Lambda(\mathcal{W}_2, \mathcal{L}_2)) = \text{dens}(\mathcal{L}_2) \theta_H(\mathcal{W}_2).$$

The assumption $\theta(\Lambda_{\text{vis},S}) > 0$ implies $\theta_H(\mathcal{W}_2) > 0$. Since $\mathcal{W}_2$ is compact and $\theta_H(\mathcal{W}_2) > 0$, the set $\Lambda_{\text{vis},S} = \Lambda(\mathcal{W}_2, \mathcal{L}_2)$ is a weak model set of maximal density.

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